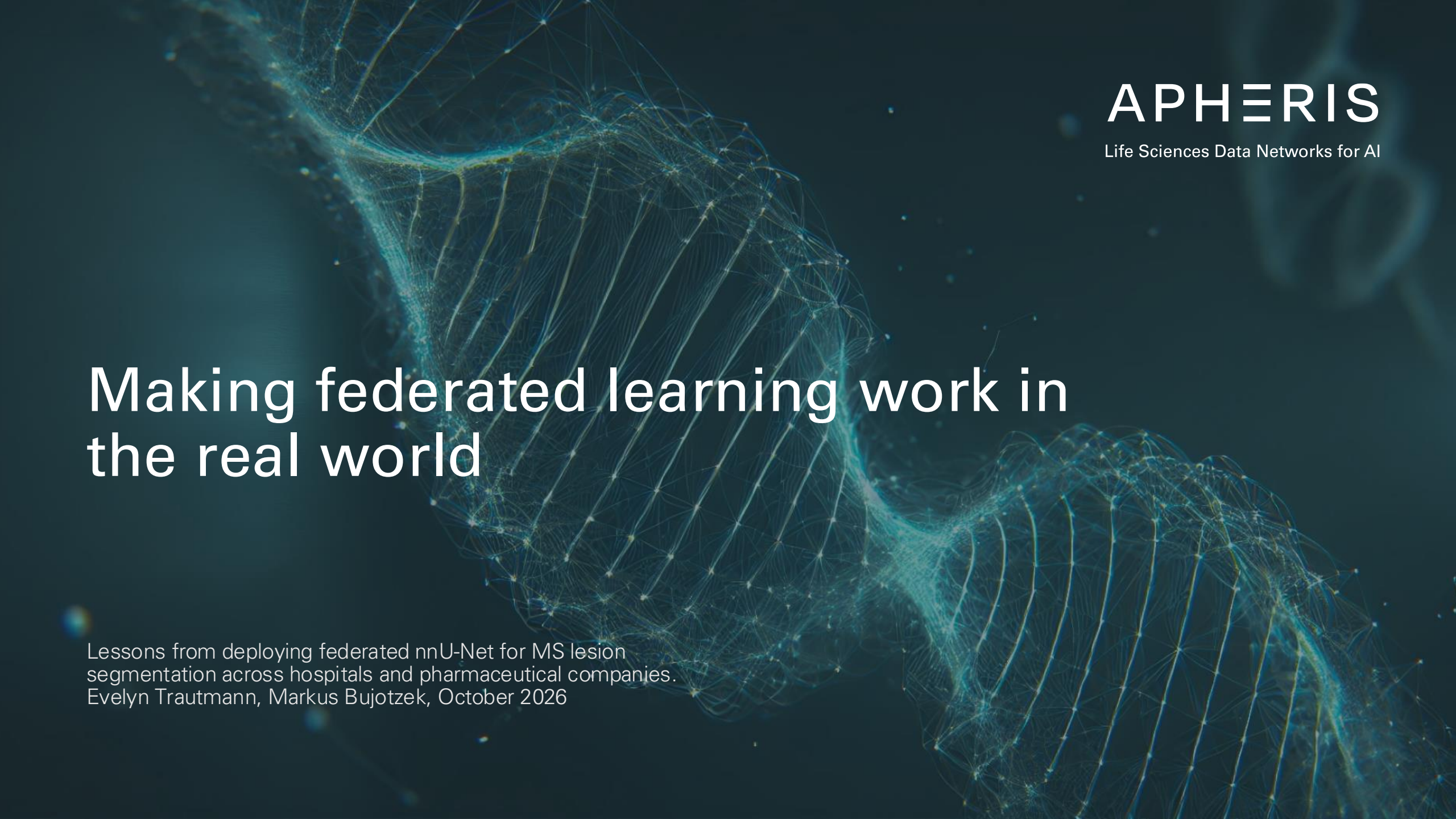


APHERIS

Life Sciences Data Networks for AI

Making federated learning work in the real world

Lessons from deploying federated nnU-Net for MS lesion segmentation across hospitals and pharmaceutical companies.

Evelyn Trautmann, Markus Bujotzek, October 2026

Segmentation quality improves when models are trained on large, heterogeneous datasets

THE CONSTRAINT

Performance is constrained by the quantity and diversity of available training data.

- **Scanners**
- **Acquisition protocols**
- **Patient populations**

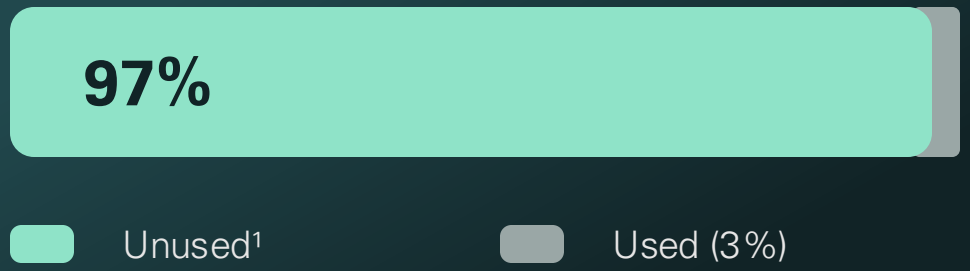
Isensee F, et al., nnU-Net Revisited, MICCAI 2024

The data access dilemma

Life sciences and health data remains inaccessible. It is distributed across organizations and cannot be shared.

"Data access in Europe is virtually impossible."
Marisa Co, VP Real World Evidence, BMS

Life sciences and health data that goes **unused**



3 key barriers: regulation, security concerns, IP considerations

1. World Economic Forum Insight Report 2024

The ecosystem is responding to the data access dilemma

Small molecules

MELLODDY
10 pharma, 21M+ compounds

Lilly TuneLab
Lilly models, opened to biotechs

ADMET Network
Five pharma and biotechs
APHERIS

Biologics

AI Structural Biology Network
Nine of the top 20 pharma
APHERIS

FAITE
AbbVie, Amgen, AZ, J&J, UCB

Antibody Developability
With Ginkgo Datapoints
APHERIS

Clinical Data & Imaging

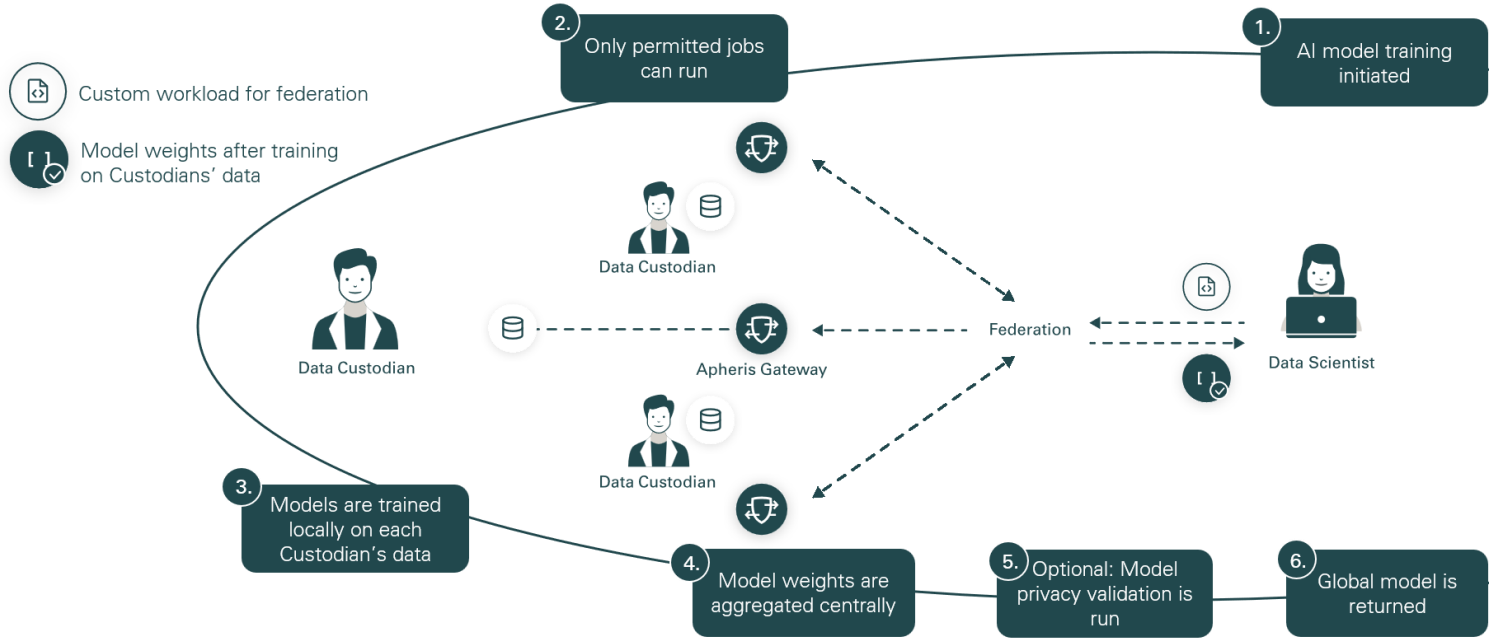
EHDEN
European real-world evidence

FeTS
71 sites, 6,314 cases

RACOON
Unifies all 38 university hospitals in Germany

The ecosystem answered with federation: Data stays where it is generated.

Federated data networks enable secure data collaboration across institutions



- Training happens where your data lives
- By definition, the data does not move
- Only model updates are exchanged
- All computational jobs are logged

THE ANSWER: Collaboration without data sharing at scale, with enhanced compliance.

Apheris hosts the industry's largest federated data networks for drug discovery

Apheris Foundry

Modules and workflows for drug programs

- Run
- Customize
- Integrate
- Close the loop

NETWORK

AI Structural Biology (AISB) Network

abbvie · astex pharmaceuticals · AstraZeneca · Boehringer Ingelheim · Bristol Myers Squibb · Genentech (A Member of the Roche Group) · Johnson&Johnson · sanofi · Takeda

NETWORK

ADMET Network

Lundbeck · ORION PHARMA · Recursion · SERVIER

+2 additional partners in onboarding

NETWORK

Virtual Cell Network

SANDBOXAQ + PHARMA AND BIOTECH MEMBERS

NETWORK

Antibody Developability Network

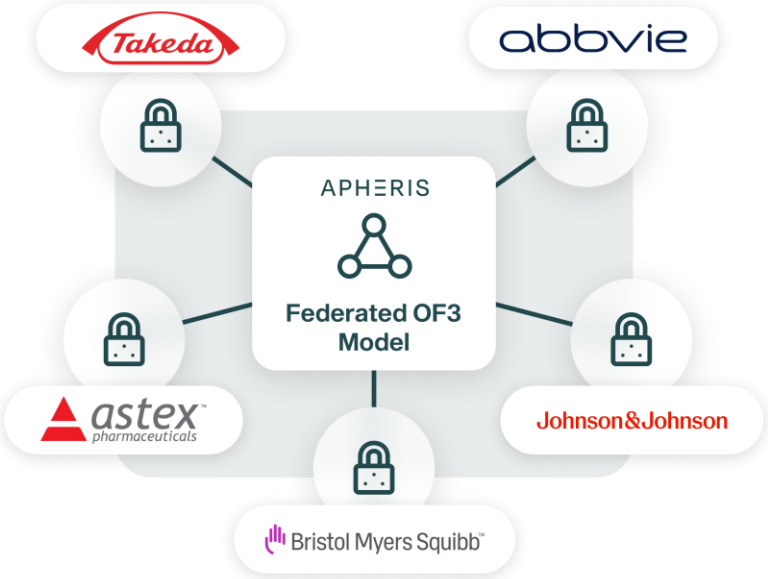
GINKGO DATAPOINTS · 4 LARGE PHARMA MEMBERS

ECOSYSTEM PARTNERS
Computing, tooling & model integration

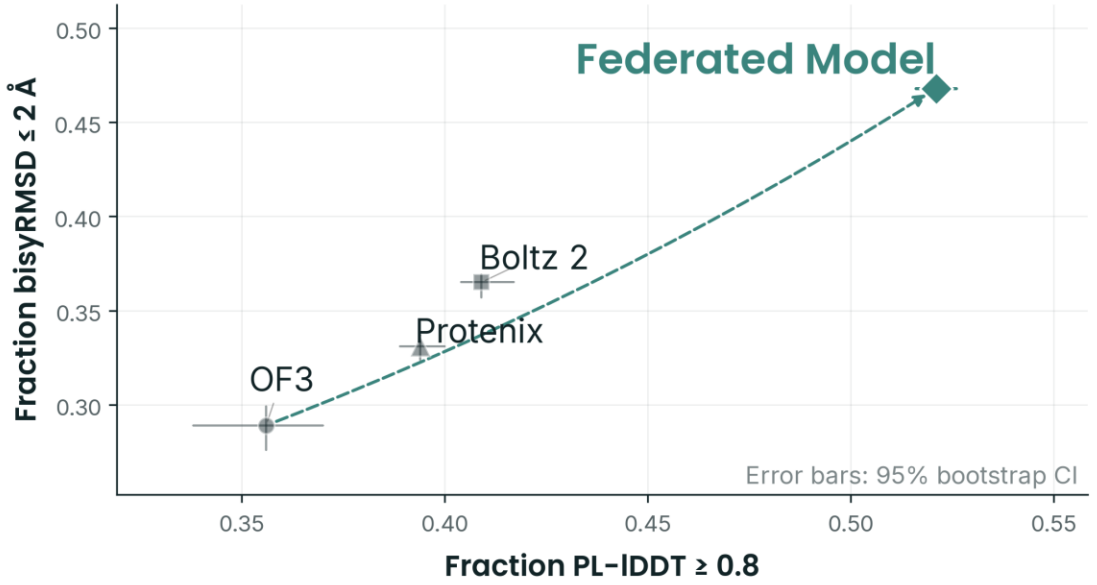
amazon Bio Discovery · NVIDIA · Tamarind Bio

Access to diverse, proprietary data fundamentally changes what models can learn. We demonstrated that data is the differentiator

One federated model



Federated training beats every public baseline



A Real-World Federated Network

MS lesion segmentation, across several clinical sites.



TYPE Brief Research Report
PUBLISHED 10 September 2025
DOI 10.3389/fneur.2025.1620469

Federated learning for lesion segmentation in multiple sclerosis: a real-world multi-center feasibility study

Sarah Hindawi[1]*, Bartlomiej Szubstarski[2], Eric Boernert[3], Björn Tackenberg[3] and Jens Wuerfel[3]

[1]Hoffmann-La Roche Limited, Mississauga, ON, Canada, [2]Roche Polska Sp. z o.o., Warsaw, Poland, [3]F. Hoffmann-La Roche AG, Basel, Switzerland

Multiple sclerosis (MS) is a chronic neuroinflammatory disease driven by immune-mediated central nervous system damage, often leading to progressive disability. Accurate segmentation of MS lesions on MRI is crucial for monitoring disease and treatment efficacy; however, manual segmentation remains time-consuming and prone to variability. While deep learning has advanced automated segmentation, robust performance benefits from large-scale, diverse datasets, yet data pooling is restricted by privacy regulations and clinical performance remains challenged by inter-site heterogeneity. In this proof-of-concept work, we aim to apply and adopt Federated Learning (FL) in a real-world hospital setting. We assessed FL for MS lesion segmentation using the self-configuring nnU-Net model, leveraging 512 MRI cases from three sites without sharing raw patient data. The federated model achieved Dice scores ranging from 0.66 to 0.80 across held-out test sets. While performance varied across sites, reflecting data heterogeneity, the study demonstrates the potential of FL as a scalable and secure paradigm for advancing automated MS analysis in distributed clinical environments. This work supports adopting secure, collaborative AI in neuroimaging, offering utility for privacy-sensitive clinical research and a starting point for medical AI development, bridging the gap between model generalizability and regulatory compliance.

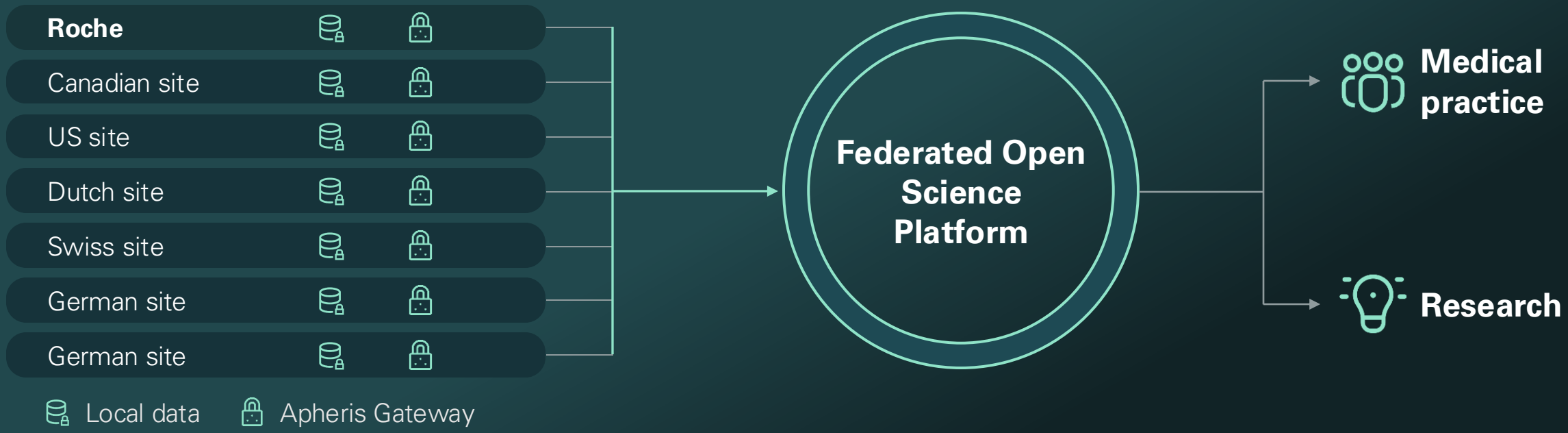
KEYWORDS

federated learning, MRI lesion segmentation, privacy-preserving AI, distributed deep learning, multi-site training

1 Introduction

Multiple sclerosis is a chronic autoimmune disorder of the central nervous system (CNS) and is a leading cause of non-traumatic neurological disability among young adults (1). MS

Addressing collaborative research challenges using federated architectures



Seven sites: Data stays local, insight is shared.

1. Rieke N, et al. NPJ Digit Med 2020;3:119; 2. Oh J, et al. Presented at ECTRIMS 2023 (Poster 1181).

Multi-center training on heterogeneous clinical MRI data for MS lesion segmentation

3

clinical sites

149, 325 and 38 cases. Heterogeneous imaging protocols and scanner types.

512

MRI cases

380 used for training. Automated lesion and lesion volume detection.

0

images moved

Kaapana for the imaging platform, Apheris for governance and orchestration.

Hindawi S, et al., "Federated learning for lesion segmentation in multiple sclerosis", 2025, Frontiers in Neurology 16

nnU-Net configures itself from the whole dataset. In federation, nobody has the whole dataset.

Why we chose nnU-Net

The reference framework for medical segmentation; transformer and Mamba methods largely fail to beat a well-configured U-Net under rigorous validation.

What federation changes

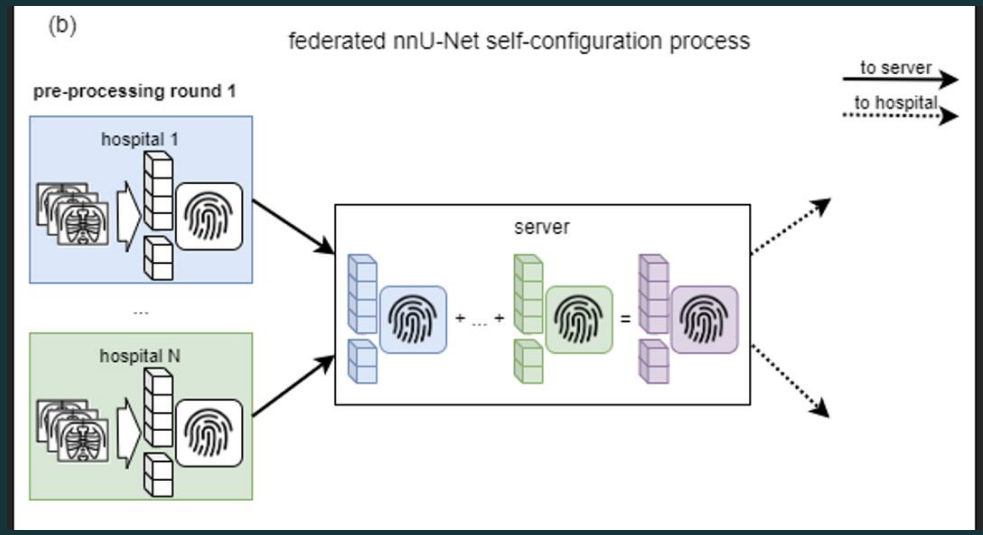
Before training, nnU-Net reads statistics across the whole dataset and from them selects the preprocessing, patch size, topology and batch size. In a federated set-up, no party sees all the cases, so that step cannot run as designed.
Run federated pre-step for fingerprinting.

The trade: You commit to one configuration for every site before you can see how it performs.

nnU-Net under the hood

Dataset statistics (fingerprint)

- **Image sizes** and spacing (voxel dimensions) for every case
- **Intensity distributions** (mean, std, percentiles) per-channel for CT/MR-like modalities
- Number of **modalities/channels**
- Class/**label distribution** (how many foreground classes, how imbalanced they are)
- Number of **training cases**



Federated nnU-Net setup



Standard nnU-Net

- Fixed parameters (like architecture template)
- Rule based heuristics for pre-processing and network topology (network depth, pooling ops, batch size, etc)
- Empirical parameters (configuration or ensemble of configurations)

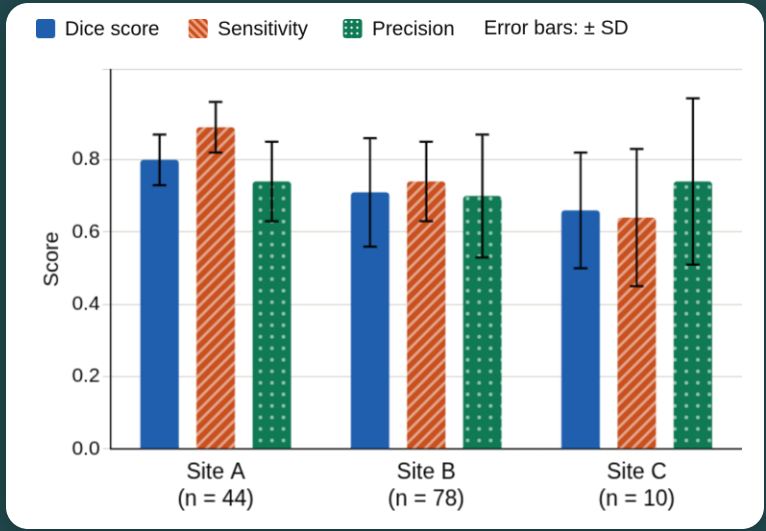
Federated Setup

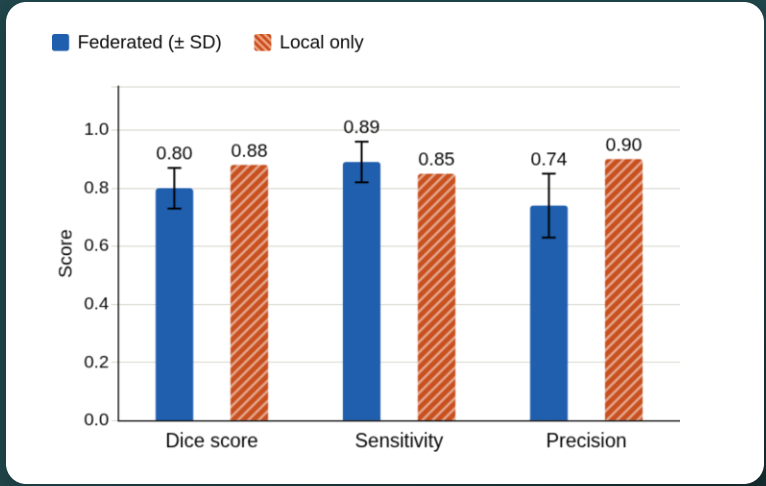
- Select one configuration before-hand, keep rule based heuristics based on initial federated fingerprinting round
- Trained 50 communication rounds
- Returns weights only, no raw image leaves the site.

No expensive hyperparameter search needed

nnU-Net uses rule-based automated self-configuration derived from a dataset fingerprint

Federated model: more sensitive, less precise, and site-dependent





First real-world (not simulated) FL deployment for MS lesion segmentation across independent hospitals

Performance varied significantly by site ($p = 3.05 \times 10^{-5}$), reflecting scanner and protocol heterogeneity rather than a flaw in the method.

Sensitivity-precision trade-off is clinically relevant: catching more lesions is valuable, but more false positives can add clinician workload.

Demonstrates technical and regulatory feasibility: a starting point for scaling FL to larger, multi-modal, longitudinal MS studies.

Federation proved practical in both studies

Federated Analytics

- Deeper understanding of MS progression patterns and societal value of MS treatment as measured in clinical practice
- Includes 25k+ people with MS across 7 sites

Key learnings

- Development within the "sciences" triangle
- Key to find the right balance between privacy and utility of the data science workflow (Layered approach).
- Overinvest in understanding data heterogeneity (both biases and standards)

Federated Learning

- Multi-center training on heterogeneous clinical MRI data for MS lesion segmentation (automated lesion and lesion volume detection)
- Quantitative and qualitative model evaluation

Key learnings

- FL for multi-center training on clinical MRI data is very practical
- FL enhances global model performance by improving lesion detection and reduces overfitting to site-specific features

MS: Multiple Sclerosis; MRI: Magnetic Resonance Imaging.
Based on the speaker's personal experience and/or opinion

But, real-world implementations come with challenges

The hard parts were governance, preprocessing and operations

The potential of AI depends on our ability to make data available, interoperable and to provide secure and governed access



Invest in strong governance

- Clear R&R with strong program management are key to (quality) data availability
- Clearly defined project, accountability (Contracting), study protocol and SAP
- Collaborate in partnership



Run remote pre-processing pipelines

- Define and deploy on site data pre-processing pipelines
- Deal with heterogeneous data and biases by deploying tailored preprocessing jobs to each site – requires collaboration across sites and “sciences”



Define Common Data Model and metadata repository

- Prerequisite for sites to map data to a common data model (CDM)
- CDM is not enough!
- GenAI is revolutionizing data wrangling with large impact on data acquisition effort and data depth/breadth

R&R; Roles and Responsibilities; SAP: Statistical Analysis Plan; AI: Artificial Intelligence; CDM: Common Data Model
Based on the speaker’s personal experience and/or opinion

Building trust & ensuring compliance in federated collaborations is key

Regulatory requirement	Solution approach
Data residency	Federated infrastructure: by definition, the data does not move.
Aligned with purpose	Computational governance: control which user, which data, which computation.
Anonymization	Jobs are validated and inspected. Raw data does not leave the Gateway.
Traceability	Computational governance: all computational jobs are logged.

- Contracting agreements between multiple parties are complex
- Addressing compliance requirements is difficult
- Federation distributes data protection responsibilities across multiple partners. This can lead to insufficient accountability and governance risks

Accountability is the open issue: Federation distributes data protection duties across partners.

trust.apheris.com; dsgvo-gesetz.de

The next gains come from the data no model has seen

Public models are strongest where public data is dense

Data-rich, on-distribution cases are largely solved.

The gap sits at the frontier: new biology, unusual populations, atypical acquisition.

The data that closes that gap sits with custodians

Three hospitals held 512 MS cases that no public model has seen.

The same pattern holds across life sciences use cases.

Federation is what unlocks it

Federated learning enables computational access to data without the need to move it.

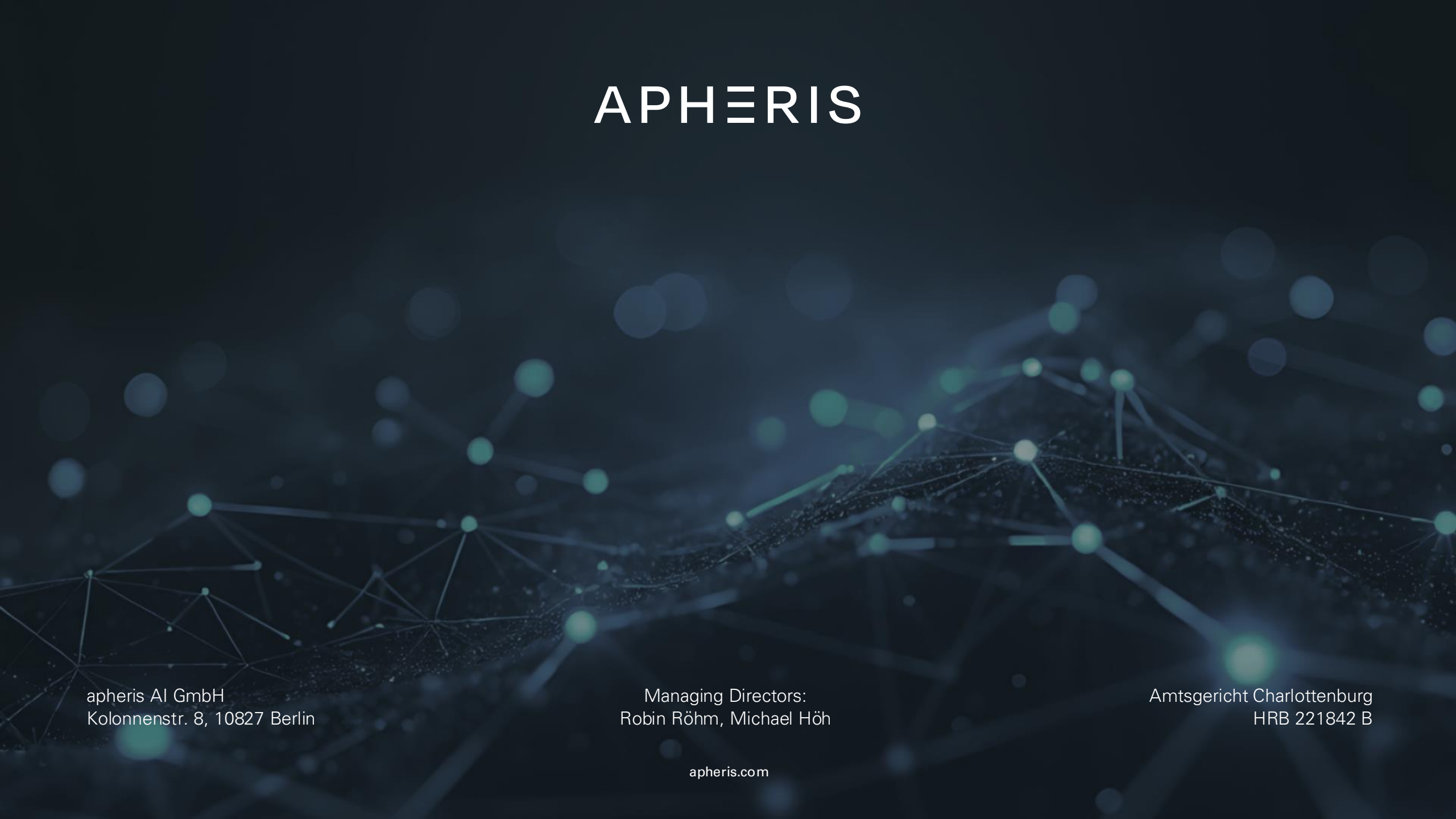
Each member added improves the model for everyone already inside.

The pattern: The thinnest, most unusual data is where federation adds the most.

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